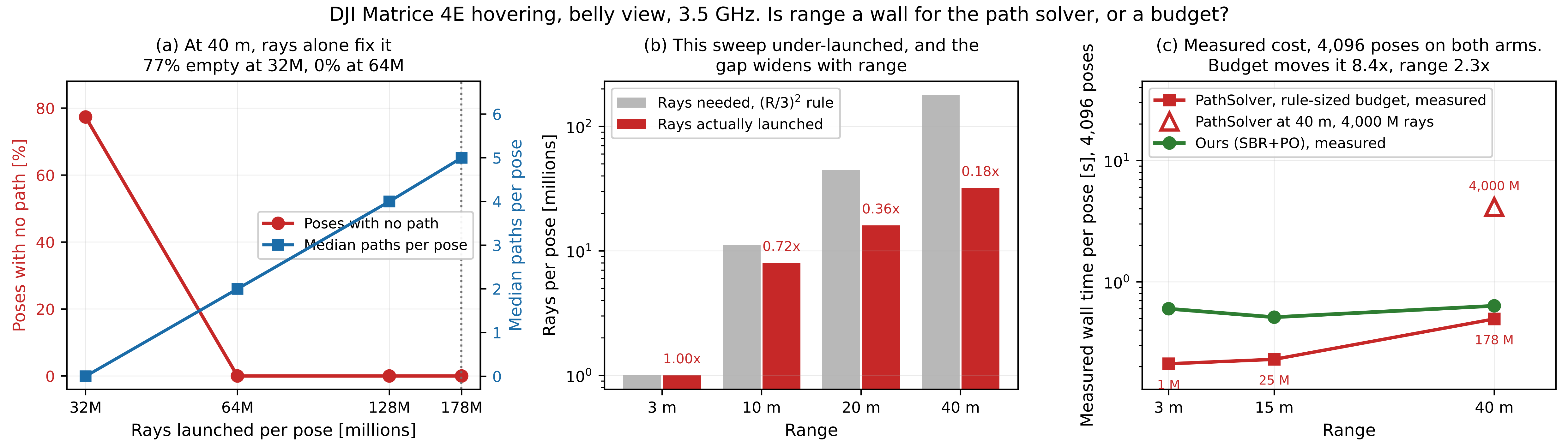

(a) Range, geometry, poses and rotor speeds are all held fixed at 40 m. Only the ray count changes, and the empty poses vanish, so the collapse at that range is a budget shortfall rather than a property of the solver. (b) The grey bars are the rule, rays scaled as (R/3) squared from the 3 m case; the red bars are what this sweep launched. Its wall time is flat across range (0.167 s per pose at 3 m down to 0.147 s at 40 m, 512 poses) because the budget stayed near the 3 m value. (c) Measurement only, at rule-sized budgets; the rule itself is panel (b). Both arms run the same 4,096 poses, same airframe, azimuth 0 and elevation -15 deg, 3.5 GHz, 19.7 kHz slow time. PathSolver budgets follow the panel (b) rule (1 M at 3 m, 25 M at 15 m, 178 M at 40 m), while ours is one frozen ray grid re-solved per range with spherical-wave phase. At short range on the minimum budget PathSolver costs less per pose, and cost crosses over as range or budget grows. Ledgers: report07_ray_budget_test.json, report07_sionna_ranges.json, report07_three_engine_ranges.json, report07_range40_raybudget.json, deck_ours_by_range.json.